

PORT TOWNSEND HARBOR, WA, USA

WMO: 994025

Lat: 48.1110N

Lon: 122.7600W

Elev: 5

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -1.9	(c) 0.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.5	(k) 8.0	(l) 12.6	(m) 8.3	(n) 3.3	(o) 310	(p) 0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.0	(c) 24.9	(d) N/A	(e) 22.6	(f) N/A	(g) 20.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.0	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.8	(b) 10.0	(c) 8.3	(e) -3.8	(f) 28.3	(g) 1.7	(h) 1.4	(i) -5.0	(j) 29.3	(k) -6.0	(l) 30.1	(m) -6.9	(n) 30.9	(o) -8.1	(p) 31.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	6.5	6.0	7.7	9.5	12.2	14.3	16.1	16.5	14.9	11.4	7.9	5.8		
	DBStd	4.43	2.31	2.78	2.30	1.80	2.14	1.96	1.99	1.76	1.75	1.98	2.72	2.80		
	HDD10.0	543	109	113	76	29	4	0	0	0	0	9	72	130		
	HDD18.3	2779	367	346	328	264	189	124	77	64	105	215	314	387		
	CDD10.0	818	1	1	6	15	73	128	188	200	146	52	7	1		
	CDD18.3	14	0	0	0	0	0	2	6	6	1	0	0	0		
	CDH23.3	121	0	0	0	0	1	19	46	48	7	0	0	0		
	CDH26.7	12	0	0	0	0	0	2	6	4	0	0	0	0		
(14) Wind	WSAvg	3.1	3.2	3.4	3.6	3.4	2.8	2.9	2.7	2.4	2.3	3.0	3.8	3.9		
(15) Precipitation	PrecAvg	488	57	40	43	39	39	31	22	22	28	36	63	65		
	PrecMax	618	167	97	88	87	77	80	74	71	81	113	122	134		
	PrecMin	342	6	8	7	7	8	4	0	0	2	0	16	10		
	PrecStd	76	30	20	19	17	17	19	17	19	19	22	27	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	12.5	15.5	17.8	22.3	25.8	27.5	26.9	24.2	18.8	14.5	12.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	11.2	11.2	13.5	15.4	19.8	22.1	24.7	24.8	21.9	16.6	13.0	11.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	10.3	10.2	11.9	13.8	17.5	19.6	22.2	22.0	20.0	15.2	12.1	10.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	9.5	9.3	10.9	12.4	15.7	17.8	20.3	20.4	18.2	14.2	11.2	9.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	3.7	4.8	5.3	6.1	6.4	7.8	8.0	6.7	4.5	3.8	3.1		
		MCDBR	4.2	4.5	6.4	7.5	9.4	10.2	11.8	11.1	9.9	6.0	4.2	3.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.316	0.327	0.348	0.374	0.357	0.347	0.351	0.339	0.328	0.323	0.317		
	taud		2.517	2.551	2.536	2.466	2.411	2.480	2.539	2.526	2.555	2.575	2.525	2.478		
	Ebn at Noon		760	855	897	905	889	905	910	892	873	827	748	709		
	Edh at Noon		62	73	86	102	112	105	98	95	84	71	61	57		
(44) All-Sky Solar Radiation	RadAvg		0.95	1.83	2.82	4.22	5.38	5.88	6.47	5.48	3.94	2.18	1.13	0.77		
	RadStd		0.13	0.24	0.33	0.40	0.43	0.57	0.47	0.36	0.34	0.24	0.17	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page